

Quantum mek II

$$\vec{B} = \frac{1}{c} \int \vec{J}(\vec{x}') \frac{(\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} d^3x' = -\frac{1}{c} \int \vec{J}(\vec{x}') \times \nabla_x \frac{1}{|\vec{x} - \vec{x}'|} d^3x' = \frac{1}{c} \nabla_x \int \frac{\vec{J}(\vec{x}') d^3x'}{|\vec{x} - \vec{x}'|} = \nabla_x \underbrace{\frac{1}{c} \int \frac{\vec{J}(\vec{x}') d^3x'}{|\vec{x} - \vec{x}'|}}_{\vec{A}} \quad \vec{B} = \nabla \times \vec{A}$$

$$\nabla \cdot \vec{E} = \mu_0 \rho$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$$

$$\nabla \times \vec{B} = \frac{\mu_0}{c} \vec{J} + \frac{d\vec{E}}{dt}$$

$$\vec{E}' = -\nabla \phi' + \frac{1}{c} \frac{d\vec{A}'}{dt}$$

$$\vec{A}' = \vec{A} - \nabla f$$

$$\phi' = \phi + \frac{1}{c} \frac{df}{dt}$$

$$\vec{B}' = \nabla \times \vec{A}'$$

$$\Rightarrow \vec{B}' = \nabla \times \vec{A}' \Rightarrow \vec{B}' = \nabla \times (\vec{A} - \nabla f) \Rightarrow \vec{B}' = \nabla \times \vec{A} - \nabla \times \nabla f \stackrel{0}{=} \Rightarrow \vec{B} = \nabla \times \vec{A}$$

$$\vec{E}' = -\nabla \left(\phi + \frac{1}{c} \frac{df}{dt} \right) + \frac{1}{c} \frac{d}{dt} (\vec{A} - \nabla f) \Rightarrow \vec{E}' = -\nabla \phi + \frac{1}{c} \frac{d\vec{A}}{dt} + \frac{1}{c} \frac{d\vec{A}}{dt} - \frac{1}{c} \frac{d\vec{A}}{dt}$$

$$\vec{E}' = \boxed{-\nabla \phi + \frac{1}{c} \frac{d\vec{A}}{dt} = \vec{E}}$$

$$\nabla \cdot \left(-\nabla \phi + \frac{1}{c} \frac{d\vec{A}}{dt} \right) = \mu_0 \rho \Rightarrow -\nabla^2 \phi + \frac{1}{c} \frac{d}{dt} \nabla \cdot \vec{A} = \mu_0 \rho$$

$$\nabla \cdot \vec{A} = 0 \Rightarrow \text{Coulomb Gauge}$$

$$\Rightarrow -\nabla^2 \phi = \mu_0 \rho \Rightarrow \nabla^2 \phi = -\mu_0 \rho \quad \text{Poisson Eq.}$$

$$\nabla \times \vec{B} = \frac{\mu_0}{c} \vec{J} + \frac{d\vec{E}}{dt} \Rightarrow \nabla \times (\nabla \times \vec{A}) - \frac{d\vec{E}}{dt} = \frac{\mu_0}{c} \vec{J}$$

$$\nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A} - \frac{d}{dt} \left(-\nabla \phi + \frac{1}{c} \frac{d\vec{A}}{dt} \right) = \frac{\mu_0}{c} \vec{J}$$

$$\nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A} + \frac{d}{dt} \nabla \phi - \frac{1}{c} \frac{d^2 \vec{A}}{dt^2} = \frac{\mu_0}{c} \vec{J}$$

$$\nabla (\nabla \cdot \vec{A} - \frac{d}{dt} \phi) - \nabla^2 \vec{A} + \frac{1}{c} \frac{d^2 \vec{A}}{dt^2} = \frac{\mu_0}{c} \vec{J}$$

$$\nabla \cdot \vec{A} = + \frac{d}{dt} \phi \quad \text{Lorentz gauge}$$

$$-\nabla^2 \phi + \frac{1}{c^2} \frac{d^2 \phi}{dt^2} = \mu_0 \rho \Rightarrow -\nabla^2 \phi + \frac{1}{c^2} \frac{d^2 \phi}{dt^2} = \mu_0 \rho \Rightarrow \nabla^2 \phi - \frac{1}{c^2} \frac{d^2 \phi}{dt^2} = -\mu_0 \rho \left(\frac{c}{c} \right)$$

$$\square^2 A_\mu = -\frac{\mu_0}{c} J_\mu \quad \text{DeLambert}$$

$$A_\mu = (c\rho, \vec{J}_\mu)$$

$$H = \frac{p^2}{2m} + V \Rightarrow H = \frac{1}{2\mu} \left(p + \frac{e}{c} A \right)^2 + e\phi$$

$$\rightarrow \frac{1}{2\mu} \left(p + \frac{e}{c} A' + \nabla f \right)$$

$$\Psi = e^{i\lambda} \Psi' \Rightarrow \Psi e^{i\lambda} = \Psi'$$

$$i\hbar \frac{d\Psi}{dt} = H\Psi \Rightarrow H\Psi = ? \Rightarrow \frac{1}{2\mu} \left(p + \frac{e}{c} A \right) \left(p + \frac{e}{c} A \right) \Psi' + e\phi \Psi' \Rightarrow \frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \Psi' + e\phi \Psi'$$

$$\frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) e^{i\lambda} \Psi + e \left(\phi - \frac{1}{c} \frac{df}{dt} \right) e^{i\lambda} \Psi$$

$$\left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) e^{i\lambda} \Psi = \frac{\hbar}{i} \nabla (e^{i\lambda} \Psi) + \frac{e}{c} A e^{i\lambda} \Psi$$

$$\Rightarrow \frac{\hbar}{i} \cdot i \nabla \lambda e^{i\lambda} \Psi + \frac{\hbar}{i} e^{i\lambda} \nabla \Psi + \frac{e}{c} A e^{i\lambda} \Psi \quad \neq \text{correct}$$

$$\Rightarrow \# = \frac{\hbar}{i} \nabla \lambda e^{i\lambda} \Psi + \frac{\hbar}{i} e^{i\lambda} \nabla \Psi + \frac{e}{c} A e^{i\lambda} \Psi$$

$$\Rightarrow \frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla \lambda e^{i\lambda} \Psi + \frac{\hbar}{i} e^{i\lambda} \nabla \Psi + \frac{e}{c} A e^{i\lambda} \Psi \right)$$

$$\left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\hbar \nabla \lambda e^{i\lambda} \psi + \frac{\hbar}{i} e^{i\lambda} \nabla \psi + \frac{e}{c} A e^{i\lambda} \psi \right)$$

$$\Rightarrow e^{i\lambda} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left[\hbar \nabla \lambda \psi + \frac{\hbar}{i} \nabla \psi + \frac{e}{c} A \psi \right]$$

$$\Rightarrow \frac{e}{c} \left[A + \frac{\hbar}{e} \nabla \right]$$

$$\Rightarrow e^{i\lambda} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \cdot \left[A + \frac{\hbar}{e} \nabla + \frac{\hbar}{i} \nabla \right] \psi = i \hbar \frac{d\psi}{dt}$$

$$i \hbar \frac{d}{dt} (e^{i\lambda} \psi) = i \hbar \frac{d\lambda}{dt} e^{i\lambda} \psi + i \hbar e^{i\lambda} \frac{d\psi}{dt} - \hbar \frac{d\lambda}{dt} e^{i\lambda} \psi + i \hbar e^{i\lambda} \frac{d\psi}{dt}$$

$$\Rightarrow \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left[A + \frac{\hbar}{e} \nabla + \frac{\hbar}{i} \nabla \right] \psi + e \phi \psi - \frac{e}{c} \frac{df}{dt} \psi + \hbar \frac{d\lambda}{dt} \psi = i \hbar \cdot \frac{d\psi}{dt}$$

$$\lambda = \frac{1}{\hbar c} e f \Rightarrow \lambda = \frac{e f}{\hbar c}$$